Exam P Daily Study Plan (January 2027)

Phase 1: General Probability (Weeks 1–3)

- **Day 1 (Session 1A.1):** Sample spaces, outcomes, and events; Venn diagrams and set operations (union, intersection, complement).
- **Day 2 (Session 1A.2):** Probability as a set function and the three basic axioms of probability.
- **Day 3 (Session 1B.1):** Fundamental counting principles, permutations, and combinations.
- **Day 4 (Session 1B.2):** Advanced combinatorics problem-solving (committees, arrangements, and distribution problems).
- **Day 5 (Session 1C.1):** Mutually exclusive events and the addition rule ($P(A \cup B)$).
- **Day 6 (Session 1D.1):** Definition of independent events and calculating joint probabilities of independent events.
- **Day 7 (Session 1E.1):** Multiplication rules and complex multi-step probability calculations.
- **Day 8 (Session 1F.1):** Definition and direct calculation of conditional probabilities.
- **Day 9 (Session 1G.1):** Law of Total Probability and partitioning sample spaces.
- **Day 10 (Session 1G.2):** Bayes' Theorem applications and probability tree diagrams.

Phase 2: Univariate Random Variables (Weeks 4–7)

- **Day 11 (Session 2A.1):** Introduction to random variables, Probability Mass Functions (PMFs), Probability Density Functions (PDFs), and Cumulative Distribution Functions (CDFs).
- **Day 12 (Session 2A.2):** Discrete distributions: Binomial and Geometric distributions.
- **Day 13 (Session 2A.3):** Discrete distributions: Negative Binomial and Hypergeometric distributions.
- **Day 14 (Session 2A.4):** Discrete distributions: Poisson distribution and its properties.
- **Day 15 (Session 2A.5):** Continuous distributions: Continuous Uniform and Exponential distributions (including the memoryless property).
- **Day 16 (Session 2A.6):** Continuous distributions: Gamma and Beta distributions.
- **Day 17 (Session 2A.7):** Continuous distributions: Normal distribution and standard normal tables (Z-scores).
- **Day 18 (Session 2B.1):** Calculating conditional probabilities for univariate random variables (truncated and restricted ranges).
- **Day 19 (Session 2C.1):** Expected values, linearity of expectation, and raw/central moments.
- **Day 20 (Session 2C.2):** Finding the mode, median, and percentiles of a distribution.
- **Day 21 (Session 2D.1):** Calculating variance, standard deviation, and the coefficient of variation.
- **Day 22 (Session 2E.1):** Modeling insurance claims with ordinary/franchise deductibles, coinsurance percentages, and benefit limits.
- **Day 23 (Session 2E.2):** Incorporating external factors such as inflation into policy modifications.
- **Day 24 (Session 2F.1):** Calculating the expected value, variance, and standard deviation of the loss random variable (X) versus the payment amount random variable (Y).

Phase 3: Multivariate Random Variables (Weeks 8–10)

- **Day 25 (Session 3A.1):** Joint probability mass functions, joint probability density functions, and joint CDFs (Discrete & Continuous).
- **Day 26 (Session 3B.1):** Determining marginal and conditional distributions from joint distributions.

- **Day 27 (Session 3C.1):** Calculating joint moments and conditional/marginal expected values.
- **Day 28 (Session 3D.1):** Calculating variance and standard deviation for conditional and marginal distributions.
- **Day 29 (Session 3E.1):** Calculating covariance, the correlation coefficient, and understanding the relationship between independence and uncorrelatedness.
- **Day 30 (Session 3F.1):** Determining the joint and marginal distributions of order statistics for a set of independent random variables.
- **Day 31 (Session 3G.1):** Calculating probabilities for linear combinations of independent random variables and bivariate normal distributions.
- **Day 32 (Session 3H.1):** Calculating moments (expected value and variance) for linear combinations of independent random variables.
- **Day 33 (Session 3I.1):** Applying the Central Limit Theorem to calculate normal approximations for sums/averages of independent and identically distributed (i.i.d.) random variables.

Phase 4: Comprehensive Review & Practice Exams (Weeks 11–12 / January)

- **Day 34 (Session 4.1):** Timed Mock Exam 1 (3 hours, 30 questions under CBT conditions).
- **Day 35 (Session 4.2):** Error analysis and targeted drilling on weak topic areas identified in Mock 1.
- **Day 36 (Session 4.3):** Timed Mock Exam 2 using the SOA Online Sample Exam P and reviewing pilot question strategies.
- **Day 37 (Session 4.4):** Final formula memorization check, review of standard normal tables, and rest prior to exam day.